Assembly line Scheduling:

$$a_1[j] = a_1[j-1] + a_2[j-1]$$
$$a_2[j] = a_2[j-1] + a_1[j-1]$$

} Solve

$a_1[j] = 2^{j-1}$, $a_2[j] = 2^{j-1}$ (done by induction or substitution)

$$\sum_{j=1}^{n} a_1[j] = \sum_{j=1}^{n} 2^{j-1} = 1+2+2^2+\cdots+2^{n-1}$$

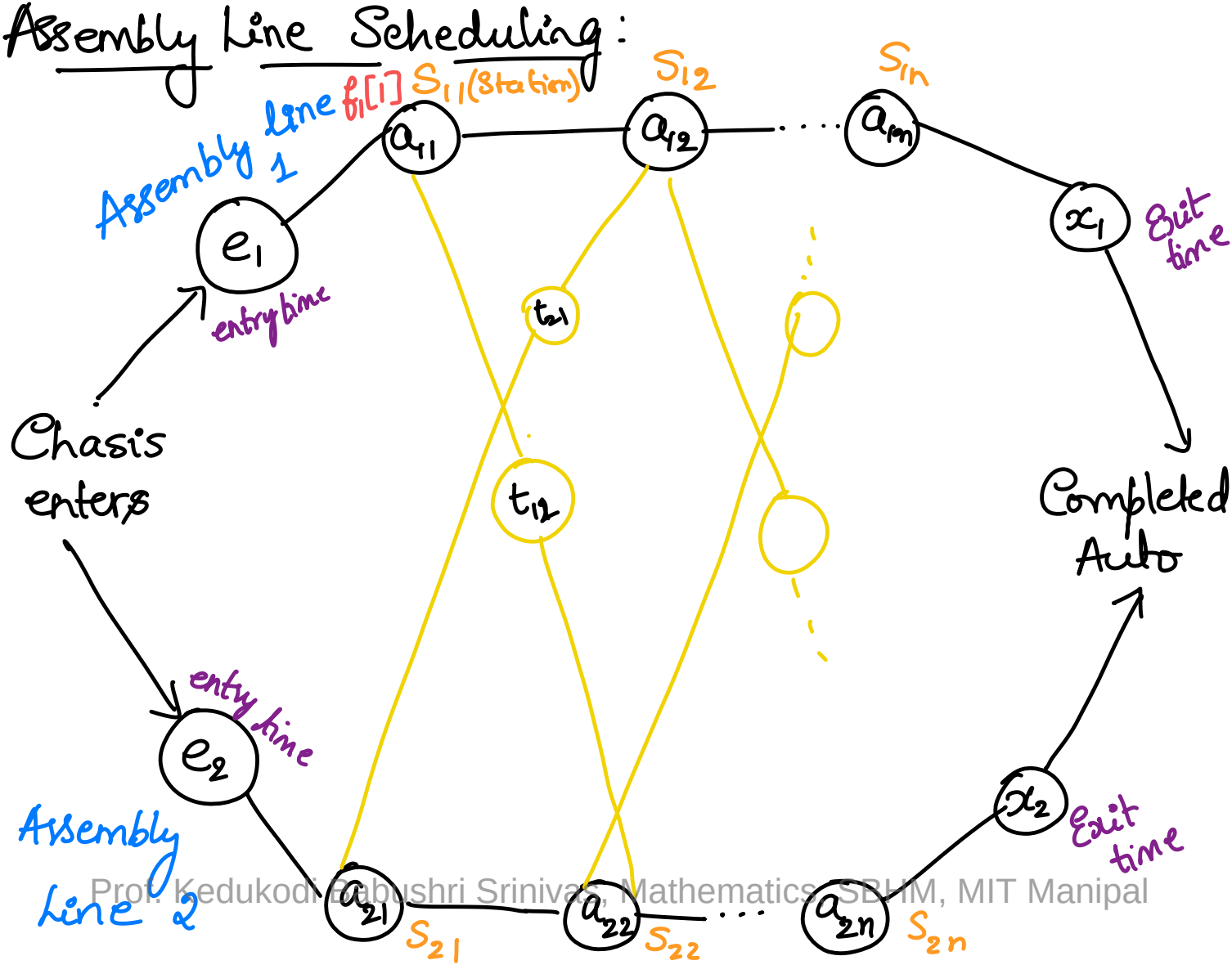
$$= \frac{2^n-1}{2-1} = \underline{\underline{2^n-1}}.$$

Similarly, $\sum_{j=1}^{n} a_2[j] = 2^n-1$

Total references made $= \sum_{j=1}^{n} a_1[j] + \sum_{j=1}^{n} a_2[j] = (2^n-1)+(2^n-1)$

$$= 2^{n+1}-2 \in \Theta(2^n)$$

Assembly Line Scheduling:



Prof. Kedukodi Babushri Srinivas, Mathematics, SBHM, MIT Manipal

Fastest way (a, t, e, x, n):

1. $f_1[1] \leftarrow e_1 + a_{11}$
2. $f_2[1] \leftarrow e_2 + a_{21}$
3. For $j \leftarrow 2$ to n
4. If $f_1[j-1] + a_{1j} \leq f_2[j-1] + t_{2j} + a_{1j}$
5. then $f_1[j] \leftarrow f_1[j-1] + a_{1j}$
6. $\ell_1[j] \leftarrow 1$ (remain in same lane)
7. else $f_1[j] \leftarrow f_2[j-1] + t_{2j} + a_{1j}$

Prof. Kedukodi Babushri Srinivas, Mathematics, SBHM, MIT Manipal

8. $l_2[j] \leftarrow 2$ (change the lane)

9. If $f_2[j-1] + a_{2j} \leq f_1[j-1] + t_{1j} + a_{2j}$

10. then $f_2[j] \leftarrow f_2[j-1] + a_{2j}$

11. $l_2[j] \leftarrow 2$

12. else $f_2[j] \leftarrow f_1[j-1] + t_{2j} + a_{2j}$

13. $l_2[j] \leftarrow 1$.

14. If $f_1[n] + x_1 \leq f_2[n] + x_2$

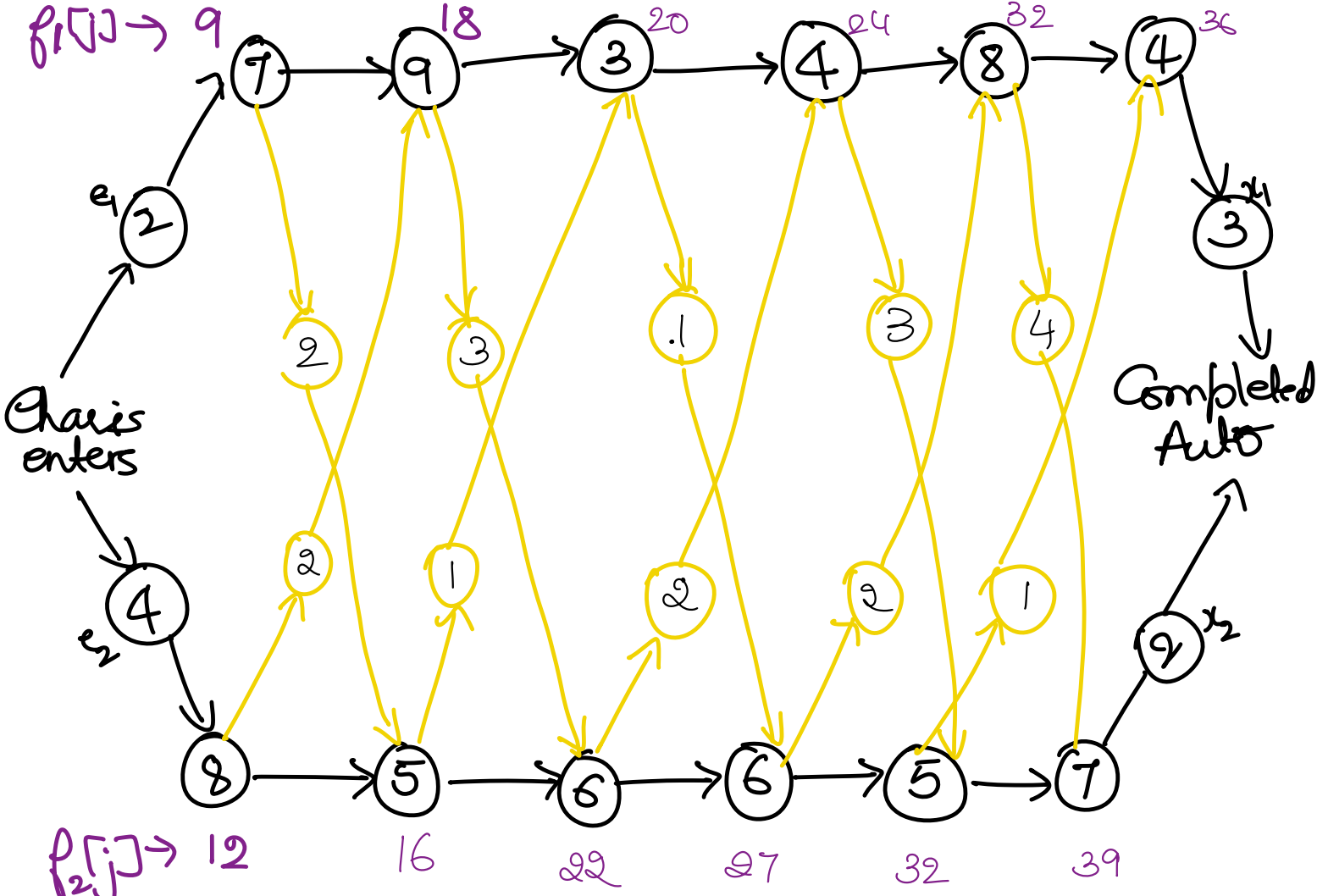
Prof. Kedukodi Babushri Srinivas, Mathematics, SBHM, MIT Manipal

15. then $f^* = f_1[n] + x_1$

16. $l^* = 1$

17. else $f^* = f_2[n] + x_1$

18. $l^* = 2$


$f_1[j] \rightarrow$ 9
18
20
24
32
36
e_1
2
7
9
3
4
8
4
x_1
3
Chairs enters
Completed Auto
2
3
1
3
4
e_2
4
2
1
2
2
1
2
x_2
8
5
6
6
5
7
$f_2[j] \rightarrow$ 12
16
22
27
32
39

$j \rightarrow$

	1	2	3	4	5	6
$f_1[j]$	7+2 9	$\min\{18, 23\}$ 18	$\min\{21, 20\}$ 20	$\min\{24, 28\}$ 24	$\min\{32, 35\}$ 32	$\min\{36, 37\}$ 36
$f_2[j]$	8+4 12	$\min\{17, 16\}$ 16	$\min\{22, \}$ 22	$\min\{28, 27\}$ 27	$\min\{32, 32\}$ 32	$\min\{39, 43\}$ 39
$l_1[j]$	1	1	2	1	1	1
$l_2[j]$	2	1	2	2	2	2

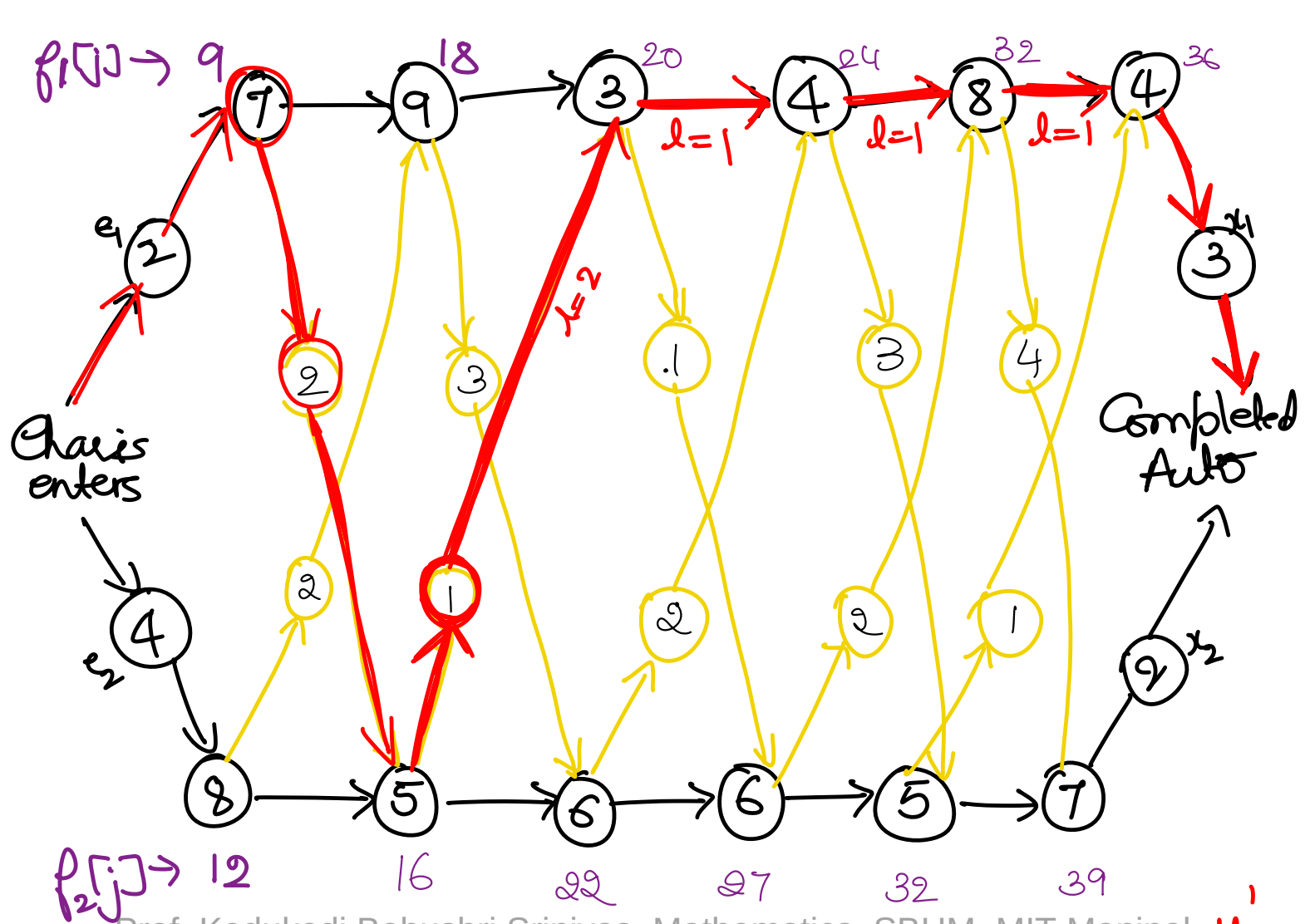
$$p^* = \min\{36 + x_1,\ 39 + x_2\}$$

$$= \min\{36 + 3,\ 39 + 2\}$$

$$= \min\{39,\ 41\}$$

$$= 39$$

$$\Rightarrow p^* = 1$$

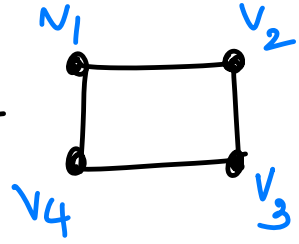

$f_1[j] \to$ 9
18
20
24
32
36
7
9
3
4
8
4
2
e_1
$l=1$
$l=1$
$l=1$
x_1
3
$l=2$
2
3
1
3
4
Chairs enters
Completed Auto
4
e_2
2
1
2
2
1
2
x_2
8
5
6
6
5
7
$f_2[j] \to$ 12
16
22
27
32
39
path 1

Graph Algorithms:

Tree → Connected graph &
no cycles (loops)

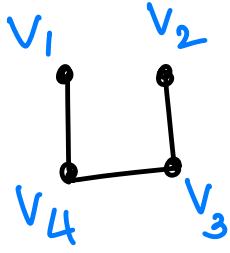
Spanning Tree of a Graph:

A tree which covers all the vertices.

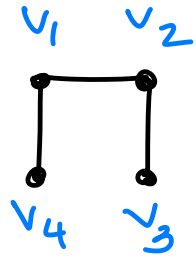
$G =$ 

Spanning Trees

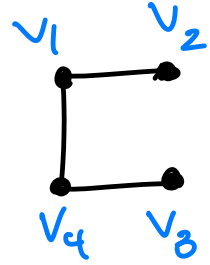
(Not unique)



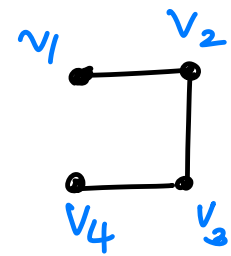
T_1



T_2



T_3



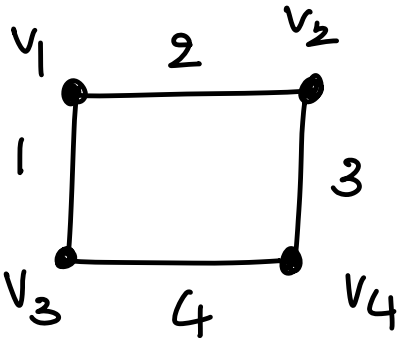
T_4

Given a graph G, there always exists a spanning tree.

Minimum Spanning Tree. (MST)

ML
Minimizati

- Weighted Graph.



T_2

T_1

$w(T_1) = 1+3+4 = 8$

$w(T_2) = 1+2+3 = 6$ ✓

T_3

$w(T_3) = 2+1+4 = 9$

T_4

$w(T_4) = 2+3+4 = 9$

T_2 is the minimum spanning tree.

Prof. Kesluku Rabushri Srinivas, Mathematics, GPHM, MIT Manipal

Brute Force:

Write all Spanning Trees

Calculate the weights

Find minimum.